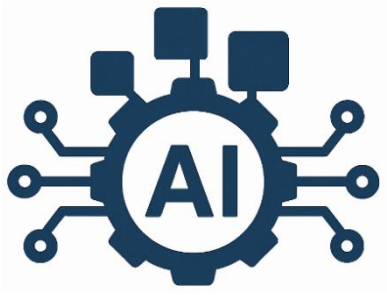
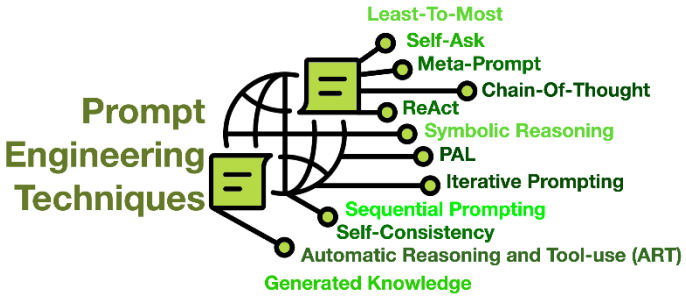
Schedule For This Weekend

Date		Topics	DDL
Dec 6	AM	• Achieving Operational Excellence and Customer Intimacy	Proposal due: **Dec 6 at 23:59**
	PM	• Ethical and Social Issues in IS	
Dec 7	AM	• Artificial Intelligence	
	PM	• Prompt Engineering	





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Achieving Operational Excellence and Customer Intimacy: Enterprise Applications

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6 Dec 2025

Kick Off – Quick Poll

Which capability matters most for competitive advantage today?

A. Operational excellence (speed, cost efficiency, reliability)
B. Customer intimacy (personalization, loyalty, experience)
C. Both equally
D. It depends on the industry

Kick Off – Pair Discussion

Turn to the person next to you and discuss:

1. A great customer experience you've had recently
   - What happened?
   - Why was it great or what made it great?
   - Was it driven by operations (e.g., speed, accuracy, availability) or customer understanding (e.g., personalization, targeted service)?
2. A frustrating experience you've had recently
   - What went wrong?
   - Was the root cause operational (e.g., inventory, delivery, coordination) or customer-related (e.g., poor service, irrelevant offers/communication)?

Be ready to share insights with the class

Kick Off – Mini Debate

Can a company deliver outstanding customer experience without strong operational excellence?

- Why or why not?
- What examples support your view? Consider industries: retail, airlines, e-commerce, banking, hospitality

Learning Objectives

- Understand
  - Enterprise resource planning (ERP) systems
  - Supply chain management (SCM) systems
  - Customer relationship management (CRM) systems
- Discuss enterprise application challenges
- Describe how enterprise applications are using AI

Enterprise systems (ERP, SCM, CRM) are strategic engines that help firms excel operationally *and* build deep customer relationships.

ERP – Definition

Enterprise resource planning (ERP) systems

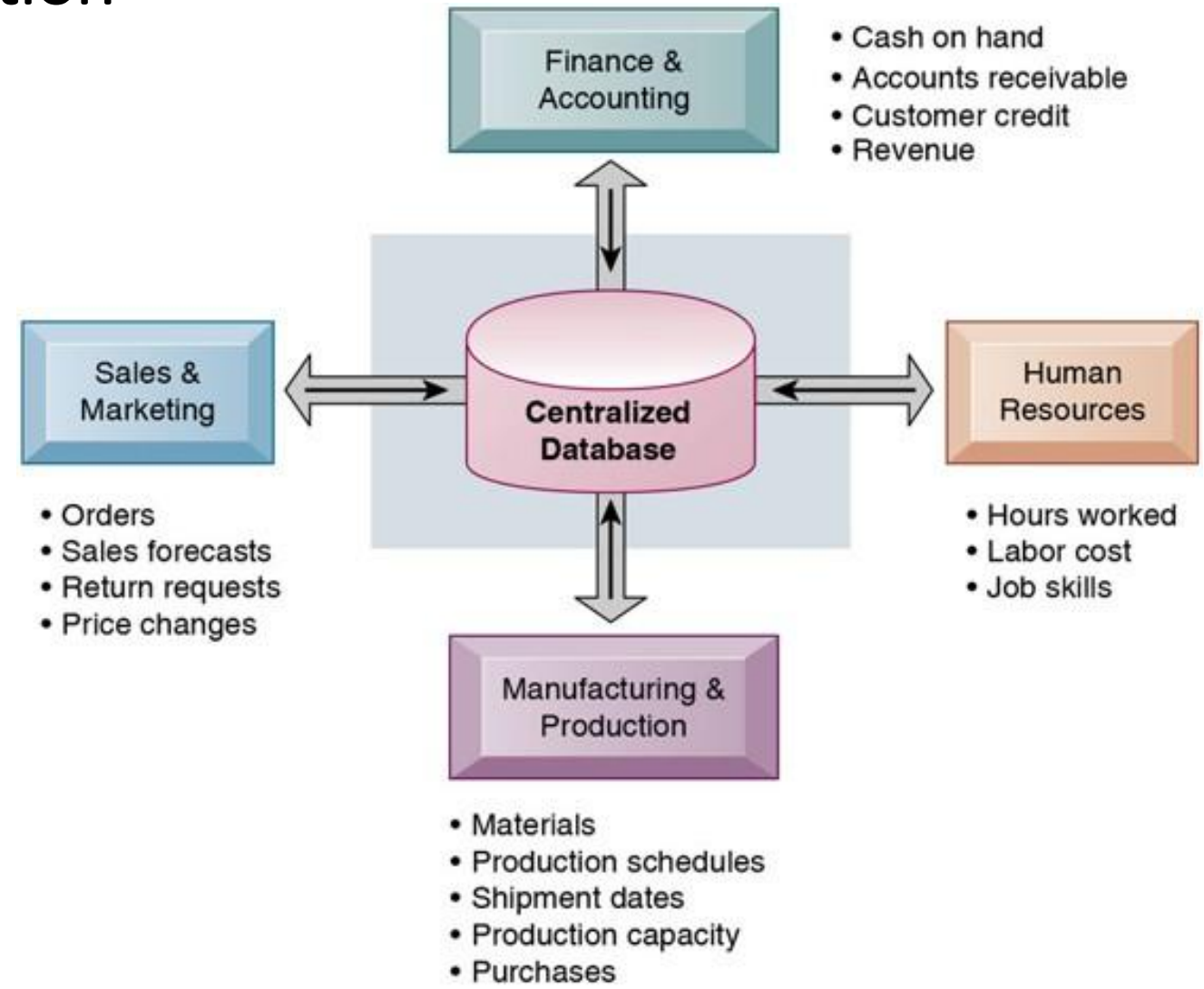
- Integrated suite of modules built on a shared central database
- Collects data across all business units and functions
- Ensures one version of the truth for internal operations
- Information entered in one area becomes instantly available to others

Why is it valuable?

Key idea

- ERP unifies the organization by integrating **processes**, **data**, and **decisions**.

ERP – Definition



ERP – Implementation

- Option 1: Build in-house ERP
  - Full control, but high cost, long development time, difficult to maintain
  - Rare today except for highly specialized industries
- Option 2: Adopt ERP from leading vendors
  - Vendors: SAP, Oracle, Microsoft, Salesforce, Blue Yonder

Example: Oracle Fusion Cloud ERP

ERP – Implementation

If adopt a vendor ERP

- Start with process–system fit
  - Select ERP modules needed
  - Map existing processes to predefined best-practice workflows
  - Use configuration tables for limited tailoring (e.g., revenue by product line/region/channel)
- Customization: A last resort
  - Rewriting ERP code is costly, risky, and can break system integration
  - Best practice: Keep customization minimal to ensure performance, upgrades, and long-term cost efficiency

ERP – Business Value

1. Drive operational excellence

- **Standardizes** processes across all locations — “one way of doing business” globally
- **Integrates** order, manufacturing, and delivery data for faster, leaner operations
- Enables firms to **produce exactly** what customers order and reduce inventory levels

Example:

Coca-Cola used SAP ERP in 200 countries to standardize processes, unlock global purchasing power, and react faster to market changes.

ERP – Business Value

2. Strengthen firm-wide decision making

- Provides real-time data on sales, production, and inventory
- Delivers accurate reporting, forecasting, and performance analytics
- Gives executives visibility into unit performance, product profitability, and cost structure

Example:

AbbVie replaced 50+ legacy systems with a single global SAP ERP, gaining:

- Standardized global processes
- Dashboards for cross-country performance
- Better root-cause analysis and more accurate reporting

Oracle AI Agent Studio for Fusion Cloud Applications: Demo

AI in ERP

Capability	Description
Digital assistant	Use AI-powered conversational chatbots to respond to customer and employee inquiries by providing support, answering questions, capturing details, and resolving issues without human interaction.
Predictive analytics	Anticipating equipment failures, recommending preventative maintenance schedules and procedures.
Anomaly detection	Analyzing ERP financial reports to identify irregularities in data entries or unusual transactions.
Improve product development and innovation	Suggest new product ideas, optimize product features and assortment.
Financial forecasting	Generating plausible financial scenarios based on potential economic, policy, or market changes.
Optimize workflows	AI can assist with almost every function of a modern supply chain, including planning, inventory and warehouse management, transaction processing, transportation, monitoring, and inspection.

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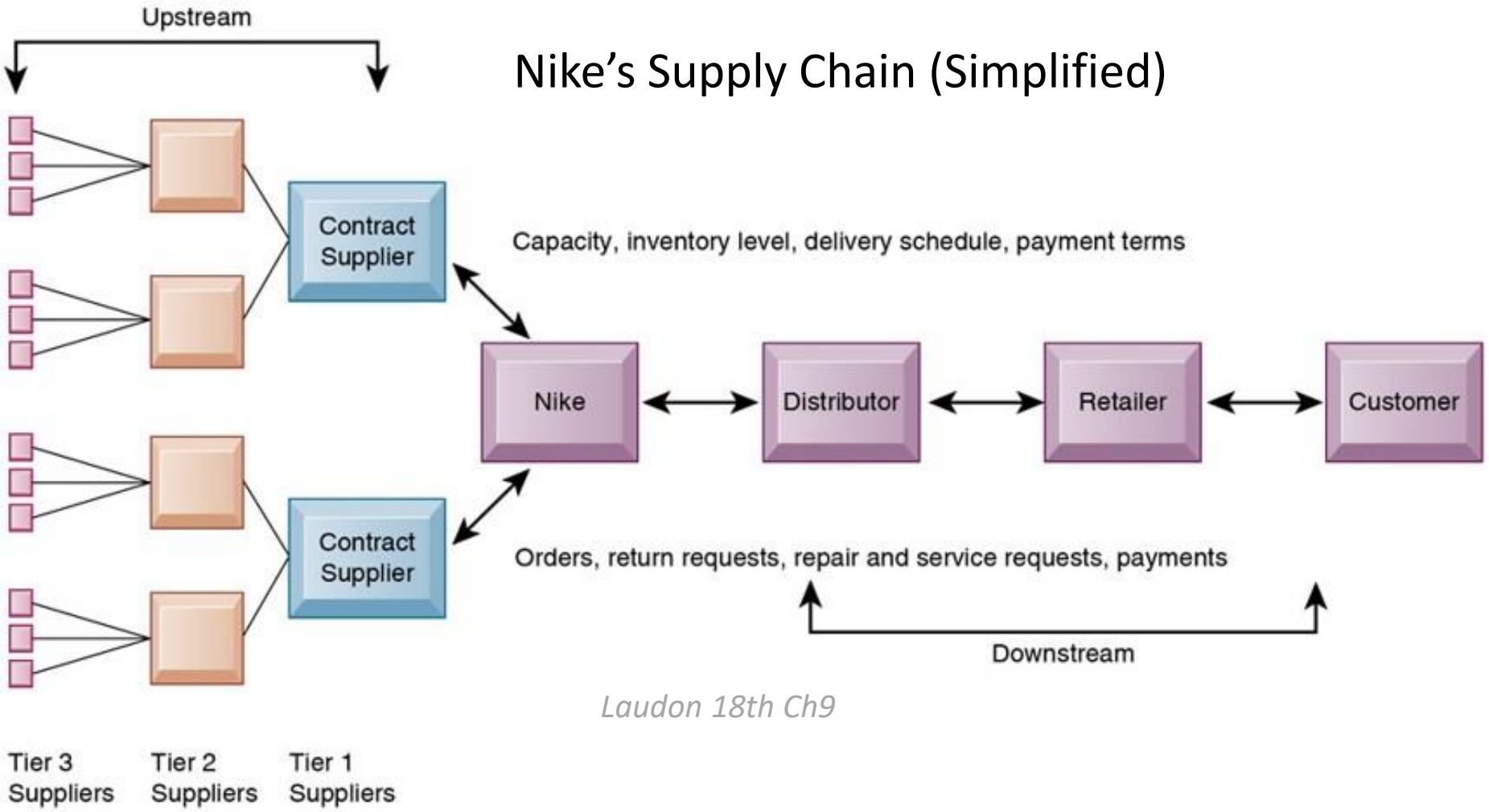
SCM – Definition

- A supply chain is a network of organizations, people, and business processes that:
  - Procure raw materials
  - Transform them into intermediate and finished products
  - Distribute those goods to customers
- Flows in the supply chain (in both directions)
  - **Materials:** raw materials → components → products
  - **Information:** order, forecasts, tracking data
  - **Payments:** invoices, credits, financial settlements

SCM – Definition



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SCM – Challenges

- Why Inefficiencies Occur
  - Inaccurate or delayed information
  - Parts shortages or excess inventory
  - Underused capacity, high logistics costs

→ Can waste up to 25% of operating costs

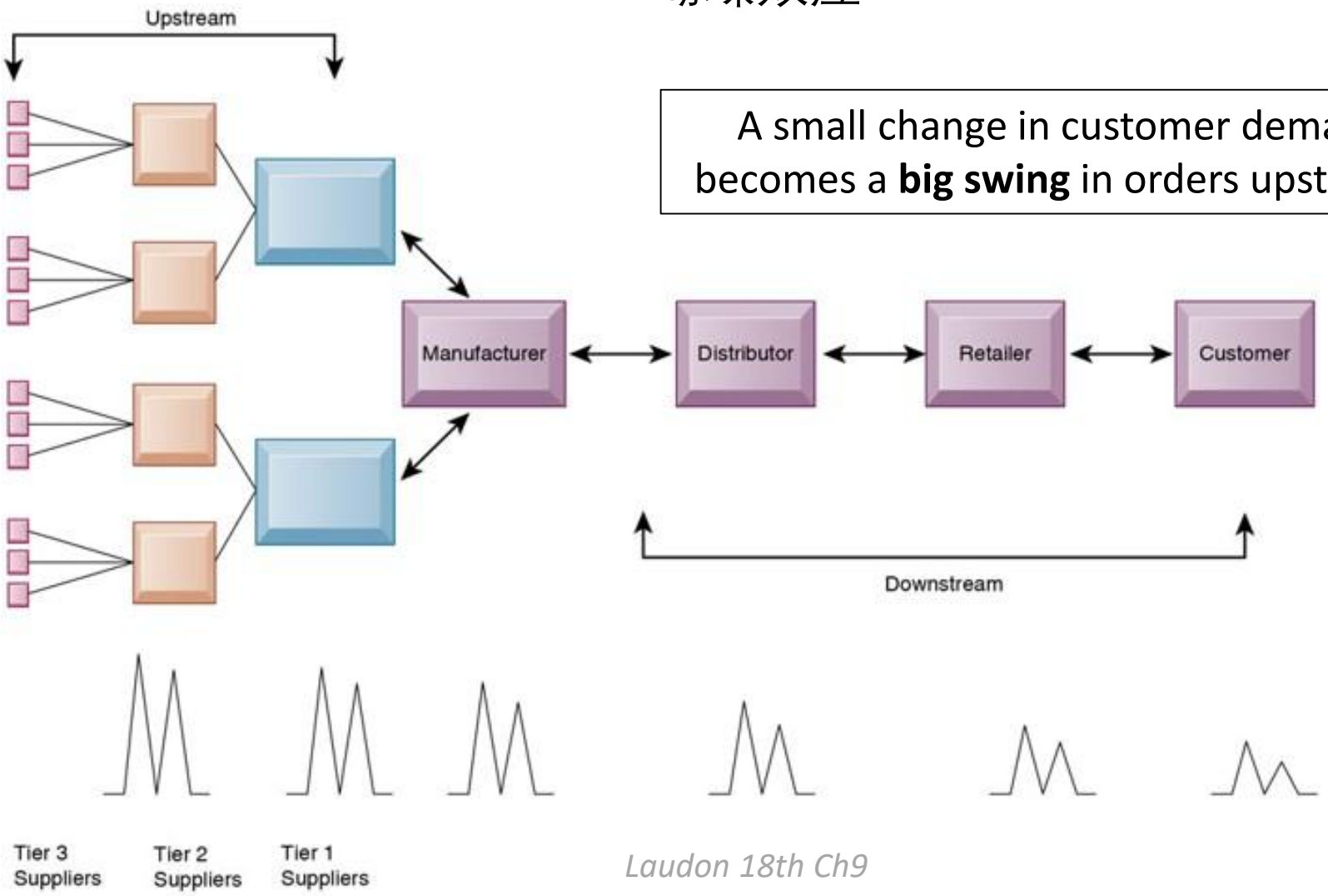
- Why Uncertainty Happens
  - Unpredictable demand
  - Late or inconsistent suppliers
  - Defects, breakdowns, disruptions

→ Firms hold safety stock — costly but needed when information is unclear.

SCM – The Bullwhip Effect

牛鞭效应
长鞭效应
蝴蝶效应

A small change in customer demand becomes a **big swing** in orders upstream.



Why it happens?

Why it matter?

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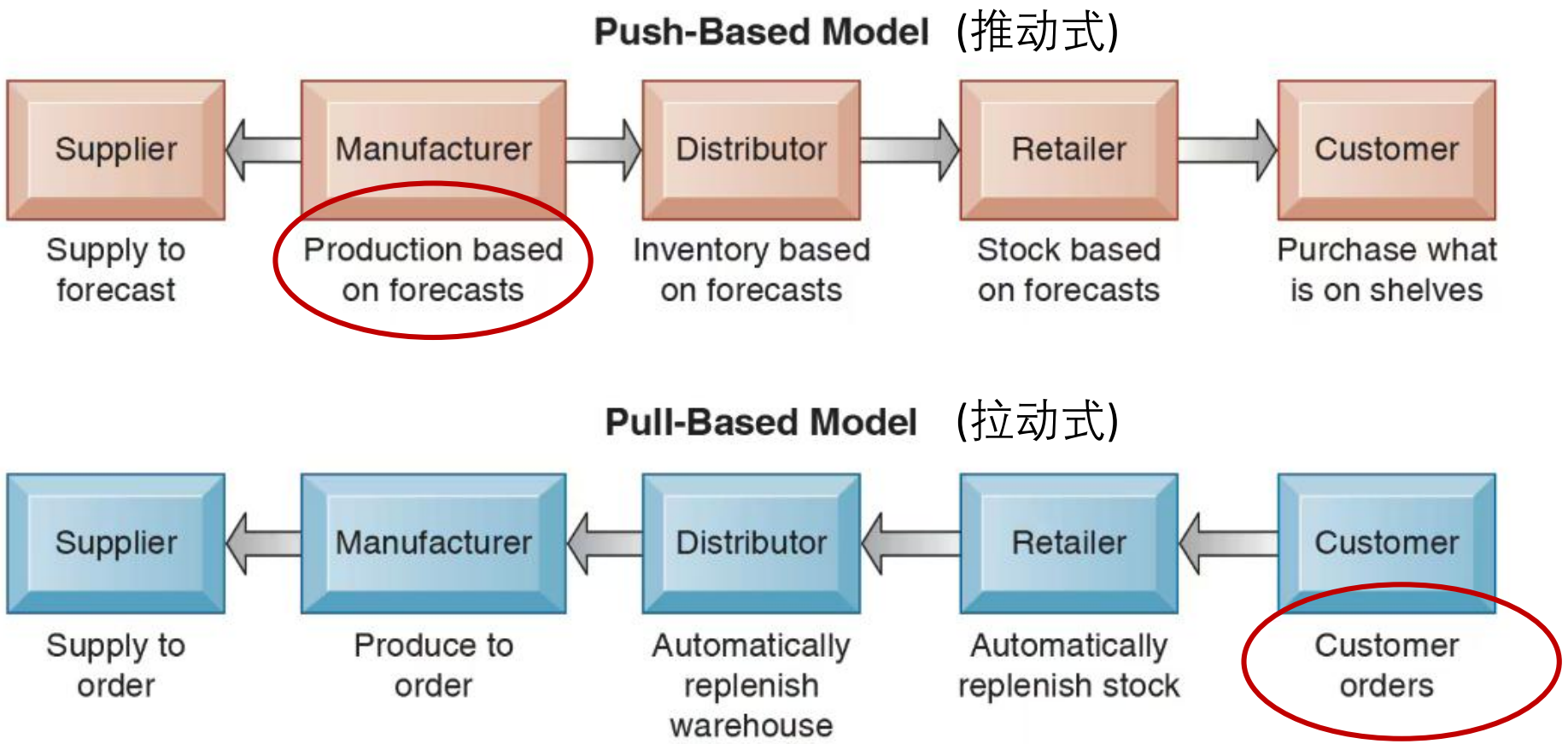
How SCM Systems Help

- Fix the Information Problem
  - Real-time visibility across suppliers, manufacturers, distributors
  - Shared data on demand, inventory, and production
- Improve Efficiency
  - Lower safety stock
  - Better forecasts
  - Faster, coordinated replenishment
- SCM systems turn a fragmented chain into a coordinated, data-driven **supply network**.

Example: Oracle Fusion Cloud Supply Chain and Manufacturing

SCM – Demand-Driven Supply Chains

Example?



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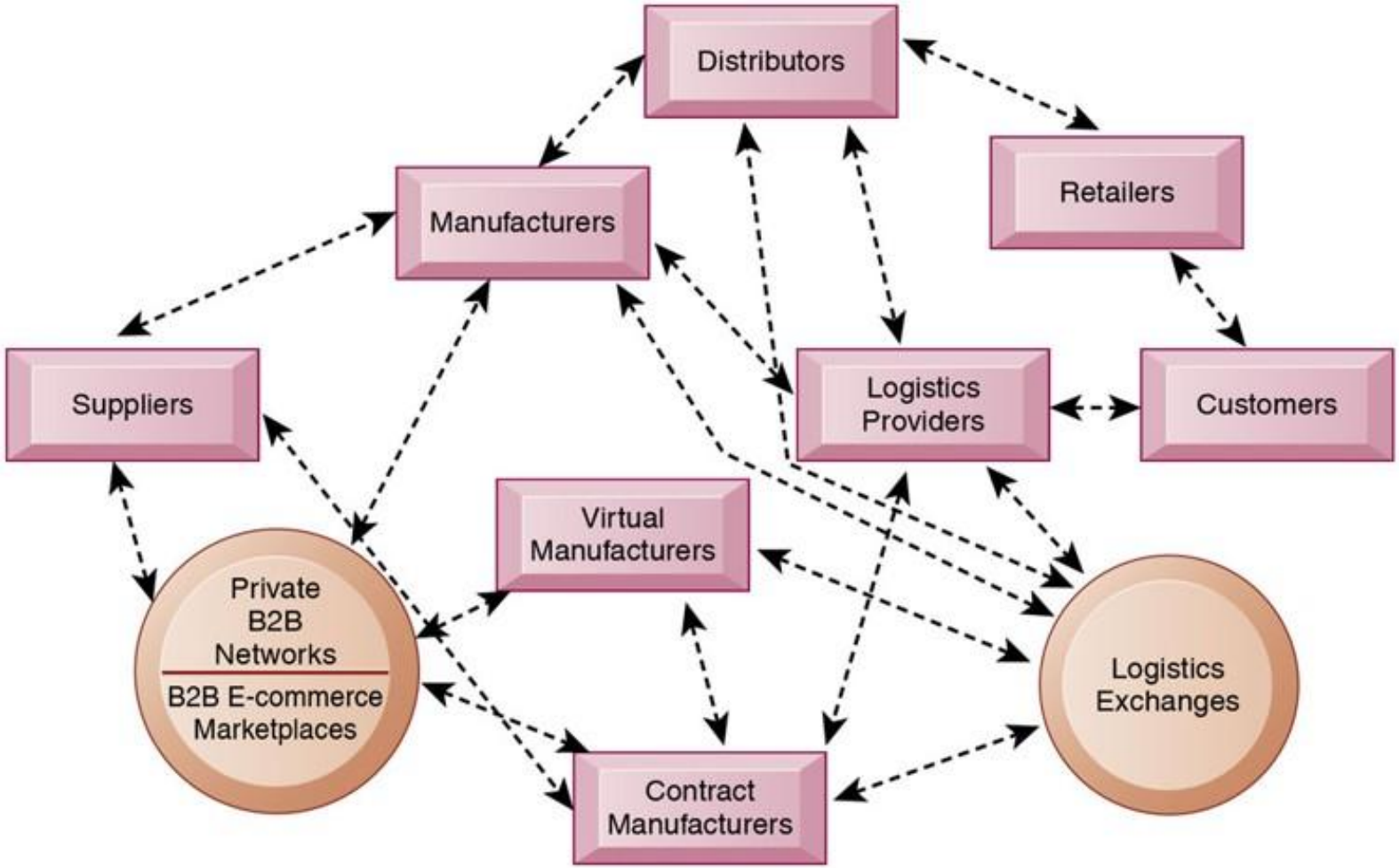
SCM – Internet-Driven Supply Chains

How should we interpret this web-like structure?



SCM – Global Challenges

- Integration across diverse systems
  - Legacy purchasing, manufacturing, and distribution systems couldn’t easily share data
  - External partners often use incompatible IT, limiting coordination
- Complexity of global operations
  - Multiple countries, time zones, and regulations
  - Longer lead times, varying standards, cultural differences
  - Exposure to tariffs, currency shifts, and trade uncertainty
- Dependence on outsourcing
  - Manufacturing spread across low-cost regions
  - Logistics handled by 3PL providers (UPS, DHL)
  - Harder to ensure quality, timing, and compliance across dispersed partners

Cvoid-19, tariff war, anti-globalization

SCM – Global Challenges

Singapore has replaced its '30 by 30' food sovereignty plan with one that sharpens focus on specific food categories, and pledged S$42M ($32M) for 11 future food projects.

Six years after announcing a strategy to lower its reliance on food imports, Singapore is overhauling its approach.

The city-state procures over 90% of its food from other countries, and only 1% of its land is set aside for agriculture, making it vulnerable to supply shocks and climate change. In its bid to boost self-sufficiency, its government launched the '30 by 30' policy in 2019, aiming to produce 30% of its food locally by the end of this decade.

Now, however, Singapore is scrapping that plan in favour of a new four-pronged strategy that targets local production of protein and fibre, import diversification, stockpiling, and global partnerships.

"We are now ready for the next chapter of our food security plan: the Singapore Food Story 2," environment minister Grace Fu said on the opening day of the Asia-Pacific Agri-Food Innovation Summit. "It replaces '30 by 39, which is an aspiration only for local production."

She added: "We can complement local production with other food strategies to achieve food resilience. We have reassessed our needs, our resources, and our capabilities. And with these considerations, we are expanding our portfolio of strategies."

AI in SCM

- Demand forecasting
  - Machine learning, predictive analytics
- Inventory management
  - AI-powered robotics, IOT sensors
- Logistics and transportation optimization
  - Route optimization algorithms, real-time tracking
- Supplier relationship management
  - AI-based supplier risk assessment, chatbots
- Quality control and predictive maintenance
  - Computer vision, AI sensors

Example: AI-Driven Supply Chain Transformation (Heineken at AIPCon 6)

Why CRM Matters

- Purpose of CRM
  - Build and maintain strong, long-lasting customer relationships
  - Understand who customers are, what they need, and how they behave
  - Make high-value customers feel recognized and rewarded
- Challenge for large firms
  - Too many customers to know **personally**
  - Many **disconnected** touch points (e.g., web, phone, email, social media, stores)
  - Sales, service, and marketing hold **fragmented** data

→ Result: No unified view of the customer.

How CRM Systems Help

Example: [What is CRM and How Does it Work?](#) (by Salesforce)

- Captures and integrates customer data from all touch points
- Consolidates and analyzes customer information
- Provides a single enterprise view for sales, service, and marketing
- Supports better service, targeted marketing, and more effective sales

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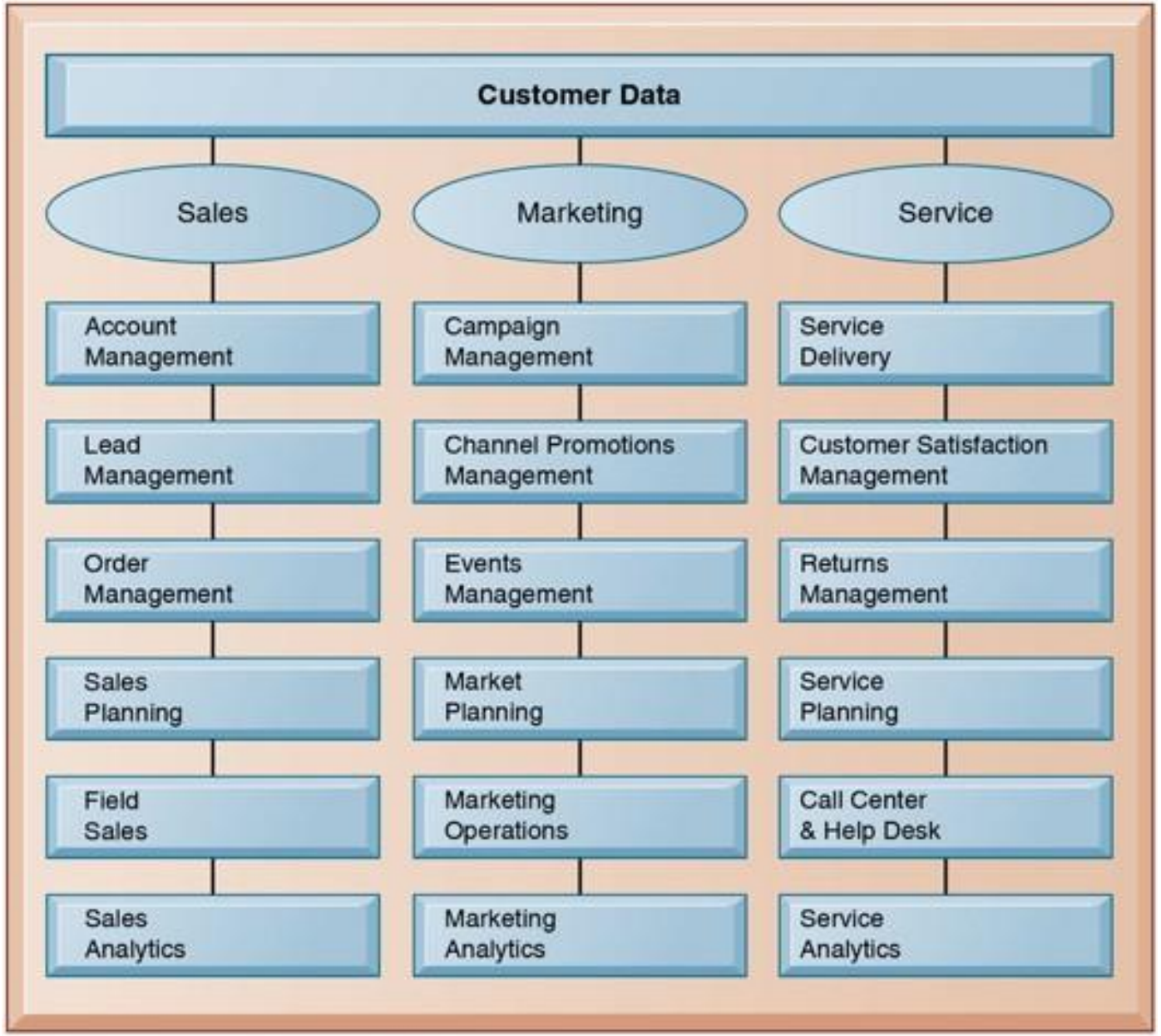
Business processes supported by CRM systems



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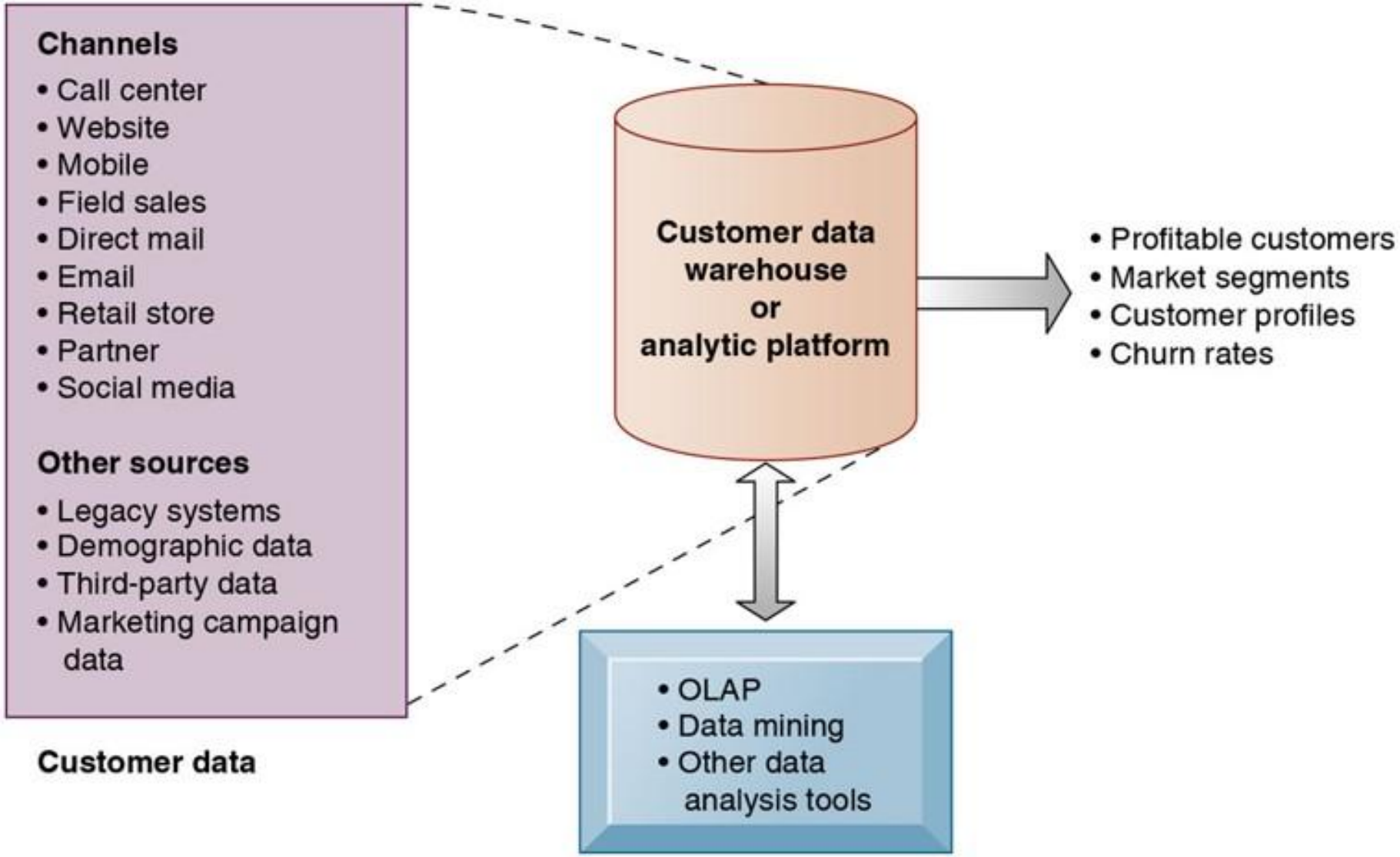
Analytical CRM



Example: 3 Ways Salesforce Uses AI with Data and CRM

AI in CRM

Capability	Description
Digital assistant agent	Use AI-powered conversational bots to respond to customer inquiries by providing support, answering questions, capturing details, and resolving issues without human interaction.
Field service routing optimization	Use predictive AI and machine learning with individual performance pattern data to optimize field service scheduling to maximize efficient use of resources and to help service teams complete more jobs per day.
Lead conversion and sales opportunity win probability	Provide sales reps with rapid identification of their best leads and sales opportunities with predictive scoring and win probabilities. Additionally, identify sales deals that are falling behind or need more attention.
Pricing optimization	Use AI-recommended pricing and discounts to strike a balance between achieving the most revenue with the highest likelihood to close.
Predictive scoring	Use machine learning to predict which leads and acts are most likely to engage and convert to buyers.

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Enterprise Application Challenges

1. High cost & implementation risk
   - Complex, expensive systems; frequent cost overruns and delays
   - Scope creep and customization inflate project complexity
2. Major organizational change
   - Requires rethinking business processes
   - Employees must learn new workflows and responsibilities
3. Data & integration issues
   - Need consistent, organization-wide data definitions
   - Data cleansing often required (especially for CRM)
   - Hard to integrate with partners across the supply chain

Enterprise Application Challenges

4. Vendor lock-In
   - Switching ERP/SCM/CRM vendors is costly and risky
   - Firms depend on vendors for upgrades and long-term support
5. Evolving solutions
   - Vendors now offer cloud-based, more flexible versions
   - Growing use of analytics, AI, and machine learning
   - Trend toward composable ERP: mixing best-of-breed modules rather than one monolithic system

Summary

- ERP – Integrate & Streamline
  - Unifies core processes and data
  - Improves efficiency and decision-making
- SCM – Coordinate & Optimize
  - Enhances visibility, reduces uncertainty
  - Builds responsive, demand-driven supply chains
- CRM – Understand & Engage
  - Creates a single view of the customer
  - Enables better service, targeting, and loyalty
- Challenges – Beyond Technology
  - High cost, implementation risk
  - Organizational change and data integration
  - Vendor dependency and evolving architectures
- AI – Transforming All Three
  - Automates processes, enhances forecasts
  - Supports real-time decisions and innovation

Quiz

What problem does the **bullwhip effect** illustrate in supply chain management?

A. Inventory always increases as demand decreases

B. Small demand changes get amplified upstream, causing inefficiencies

C. Suppliers cannot access customer data

D. Logistics providers must hold safety stock

Quiz

A company's new ERP system goes live, but employees continue using old spreadsheets, causing data inconsistency and delays. What is the **MOST likely root cause**?

A. The ERP vendor lacked enough modules

B. The company did not minimize customization

C. Organizational change management was insufficient

D. The supply chain was too global